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Lexeme CLASS

SOURCE

An entry in the vocabulary

A `Lexeme` has no string context – it's a word type, as opposed to a word token. It therefore has no part-of-speech tag, dependency parse, or lemma (if lemmatization depends on the part-of-speech tag).

Lexeme.__init__ METHOD

Create a `Lexeme` object.

NAME	DESCRIPTION
`vocab`	The parent vocabulary. **TYPE:** `Vocab`
`orth`	The orth id of the lexeme. **TYPE:** `int`

Lexeme.set_flag METHOD

Change the value of a boolean flag.

NAME	DESCRIPTION
`flag_id`	The attribute ID of the flag to set. **TYPE:** `int`
`value`	The new value of the flag. **TYPE:** `bool`

EXAMPLE

```
COOL_FLAG = nlp.vocab.add_flag(lambda text: False)
nlp.vocab["spaCy"].set_flag(COOL_FLAG, True)
```

Lexeme.check_flag METHOD

Check the value of a boolean flag.

NAME	DESCRIPTION
`flag_id`	The attribute ID of the flag to query.

EXAMPLE

```
is_my_library = lambda text: text in ["spaCy", "Thinc"]
MY_LIBRARY = nlp.vocab.add_flag(is_my_library)
```

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RETURNS	The value of the flag. **TYPE:** `bool`

Lexeme.similarity METHOD NEEDS MODEL

Compute a semantic similarity estimate. Defaults to cosine over vectors.

NAME	DESCRIPTION
other	The object to compare with. By default, accepts `Doc`, `Span`, `Token` and `Lexeme` objects. **TYPE:** `Union[Doc,Span,Token,Lexeme]`
RETURNS	A scalar similarity score. Higher is more similar. **TYPE:** `float`

EXAMPLE

```
apple = nlp.vocab["apple"]
orange = nlp.vocab["orange"]
apple_orange = apple.similarity(orange)
orange_apple = orange.similarity(apple)
assert apple_orange == orange_apple
```

Lexeme.has_vector PROPERTY NEEDS MODEL

A boolean value indicating whether a word vector is associated with the lexeme.

NAME	DESCRIPTION
RETURNS	Whether the lexeme has a vector data attached. **TYPE:** `bool`

EXAMPLE

```
apple = nlp.vocab["apple"]
assert apple.has_vector
```

Lexeme.vector PROPERTY NEEDS MODEL

A real-valued meaning representation.

NAME	DESCRIPTION
RETURNS	A 1-dimensional array representing the lexeme's vector. **TYPE:** `numpy.ndarray[ndim=1, dtype=float32]`

EXAMPLE

```
apple = nlp.vocab["apple"]
assert apple.vector.dtype == "float32"
assert apple.vector.shape == (300,)
```

Lexeme.vector_norm PROPERTY NEEDS MODEL

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NAME	DESCRIPTION
RETURNS	The L2 norm of the vector representation. **TYPE:** `float`

```
apple = nlp.vocab["apple"]
pasta = nlp.vocab["pasta"]
apple.vector_norm  # 7.1346845626831055
pasta.vector_norm  # 7.759851932525635
assert apple.vector_norm != pasta.vector_norm
```

Attributes

NAME	DESCRIPTION
`vocab`	The lexeme's vocabulary. **TYPE:** `Vocab`
`text`	Verbatim text content. **TYPE:** `str`
`orth`	ID of the verbatim text content. **TYPE:** `int`
`orth_`	Verbatim text content (identical to `Lexeme.text`). Exists mostly for consistency with the other attributes. **TYPE:** `str`
`rank`	Sequential ID of the lexeme's lexical type, used to index into tables, e.g. for word vectors. **TYPE:** `int`
`flags`	Container of the lexeme's binary flags. **TYPE:** `int`
`norm`	The lexeme's norm, i.e. a normalized form of the lexeme text. **TYPE:** `int`
`norm_`	The lexeme's norm, i.e. a normalized form of the lexeme text. **TYPE:** `str`
`lower`	Lowercase form of the word. **TYPE:** `int`
`lower_`	Lowercase form of the word. **TYPE:** `str`
`shape`	Transform of the word's string, to show orthographic features. Alphabetic characters are replaced by `x` or `X`, and numeric characters are replaced by `d`, and sequences of the same character are truncated after length 4. For example, `"Xxxx"` or `"dd"`. **TYPE:** `int`
`shape_`	Transform of the word's string, to show orthographic features. Alphabetic characters are replaced by `x` or `X`, and numeric characters are replaced by `d`, and sequences of the same character are truncated after length 4. For example, `"Xxxx"` or `"dd"`. **TYPE:** `str`
`prefix`	Length-N substring from the start of the word. Defaults to `N=1`. **TYPE:** `int`
`prefix_`	Length-N substring from the start of the word. Defaults to `N=1`. **TYPE:** `str`
`suffix`	Length-N substring from the end of the word. Defaults to `N=3`. **TYPE:** `int`

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`suffix_`	Length-N substring from the end of the word. Defaults to `N=3`. **TYPE:** `str`
`is_alpha`	Does the lexeme consist of alphabetic characters? Equivalent to `lexeme.text.isalpha()`. **TYPE:** `bool`
`is_ascii`	Does the lexeme consist of ASCII characters? Equivalent to `[any(ord(c) >= 128 for c in lexeme.text)]`. **TYPE:** `bool`
`is_digit`	Does the lexeme consist of digits? Equivalent to `lexeme.text.isdigit()`. **TYPE:** `bool`
`is_lower`	Is the lexeme in lowercase? Equivalent to `lexeme.text.islower()`. **TYPE:** `bool`
`is_upper`	Is the lexeme in uppercase? Equivalent to `lexeme.text.isupper()`. **TYPE:** `bool`
`is_title`	Is the lexeme in titlecase? Equivalent to `lexeme.text.istitle()`. **TYPE:** `bool`
`is_punct`	Is the lexeme punctuation? **TYPE:** `bool`
`is_left_punct`	Is the lexeme a left punctuation mark, e.g. `(`? **TYPE:** `bool`
`is_right_punct`	Is the lexeme a right punctuation mark, e.g. `)`? **TYPE:** `bool`
`is_space`	Does the lexeme consist of whitespace characters? Equivalent to `lexeme.text.isspace()`. **TYPE:** `bool`
`is_bracket`	Is the lexeme a bracket? **TYPE:** `bool`
`is_quote`	Is the lexeme a quotation mark? **TYPE:** `bool`
`is_currency`	Is the lexeme a currency symbol? **TYPE:** `bool`
`like_url`	Does the lexeme resemble a URL? **TYPE:** `bool`
`like_num`	Does the lexeme represent a number? e.g. “10.9”, “10”, “ten”, etc. **TYPE:** `bool`
`like_email`	Does the lexeme resemble an email address? **TYPE:** `bool`
`is_oov`	Is the lexeme out-of-vocabulary (i.e. does it not have a word vector)? **TYPE:** `bool`
`is_stop`	Is the lexeme part of a “stop list”? **TYPE:** `bool`
`lang`	Language of the parent vocabulary. **TYPE:** `int`
`lang_`	Language of the parent vocabulary. **TYPE:** `str`

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`prob`	Smoothed log probability estimate of the lexeme's word type (context-independent entry in the vocabulary).
	TYPE: `float`
`cluster`	Brown cluster ID.
	TYPE: `int`
`sentiment`	A scalar value indicating the positivity or negativity of the lexeme.
	TYPE: `float`

SUGGEST EDITS

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